Dynamic network analysis of multi-scale -omics data for protein function prediction

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Proteins play critical roles in living organisms. Unraveling their functions aids in decoding cellular processes and discovering drug targets. Traditional wet lab experiments for determining protein function are costly, time-intensive, and low-throughput, leaving less than 0.1% of protein sequences reliably annotated [1]. Thus, there has been a pressing need for computational prediction of protein function.

Various computational methods for this purpose have been developed, utilizing -omics data types at different scales/levels of biological organization, including protein sequences, protein 3-dimensional (3D) structures, and protein-protein interactions (PPIs). Sequence-based methods are popular due to large amounts of protein sequence data being available. However, protein sequences are folded into 3D structures, and it is a protein's 3D structure that directs what other proteins the given protein can interact with, and thus which function(s) it performs in a cell. Hence, 3D structural methods are even more promising than sequence methods, especially with recent availability of computationally predicted yet high-quality 3D structural data by powerful methods such as AlphaFold2. Structure-based methods involve either directly analyzing the original 3D structure or first modeling it into a protein structure network (PSN) and then analyzing the PSN; in a PSN, nodes are amino acids of a protein and edges join those amino acids that are close in the 3D structure of the protein. PSN-based approaches have often achieved greater accuracy than sequence-based and "direct" 3D structural approaches, while also typically exhibiting faster computing speeds, in tasks of protein structural classification [2] and function prediction [3]. The power of network analysis of -omics data also applies to the next level of biological organization, namely protein-protein interactions (PPIs), which can naturally be modeled by a PPI network in which nodes are proteins and edges are physical interactions between them.

Our lab recently integrated the network representations of 3D structures and PPIs into a multi-scale "network of networks" (NoNs), where nodes in a network at a higher level are themselves networks at a lower level (Fig. 1). More specifically, the level 2 network is a PPI network, which contains level 2 nodes (proteins) and level 2 edges (PPIs). Each level 2 node is a PSN, which contains level 1 nodes (amino acids) and level 1 edges (spatial proximities between the amino acids). The NoN approach performed promising in the task of protein function prediction. It outperformed single-scale approaches that rely solely on either PSN data or PPI network data in all tests on synthetic NoNs, and for approximately half of the protein functions (i.e., Gene Ontology (GO) terms) on real PSN+PPI NoN data [4]. Notably, for 30% of the GO terms, only the NoN approach made meaningful predictions, whereas the single-scale approaches achieved random accuracy.

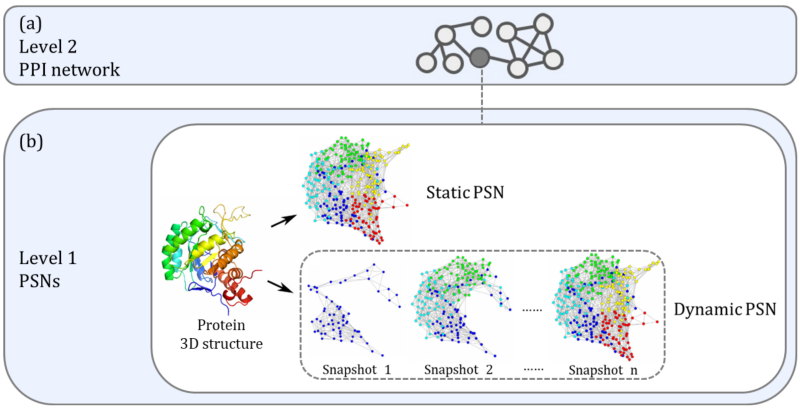

Figure 1: The NoN framework. (a) The level 2 network is a PPI network. The dotted lines join level 2 nodes (proteins) to their corresponding (b) level 1 networks (PSNs). Only one PSN is shown for simplicity, but every level 2 node can have its level 1 network. (b) Illustration of a static PSN vs. dynamic PSN generated from a protein's 3D structure. The dynamic PSN consists of multiple network snapshots approximating the dynamics of the protein folding process.

Until recently, existing PSN approaches typically represented the entire final (native) 3D structure of a protein as a static PSN. However, protein folding is a dynamic process, with certain parts of a protein folding before others. Most recently, our lab proposed to model the protein's 3D structure as a dynamic PSN, in order to capture more comprehensive information than a static PSN [5]. Intuitively, a dynamic PSN consists of multiple PSN "snapshots", each snapshot capturing the 3D sub-structure of a portion of the protein sequence. Progressively increasing portions of the sequence are captured to approximate the dynamics of the protein folding process, with each sequence portion having its own network snapshot within the dynamic PSN. The last snapshot captures the entire 3D structure of a protein, which matches its static PSN.

The (single-scale) dynamic PSN approach significantly outperformed its static PSN counterpart in the task of protein structure classification, which categorizes protein domains into predefined (e.g., CATH or SCOPe) structural classes. **In our on-going work to be presented at ISMB 2024, we evaluate whether integrating dynamic rather than static PSNs within the established multi-scale NoN framework (Fig. 1) will yield higher accuracy in the protein function prediction task.** Specifically, for a given protein, we extract graph features from its dynamic or static PSN and subsequently integrate these with node features corresponding to the same protein derived from the PPI network. We analyze a biological NoN consisting of a human PPI network (18,708 nodes, 434,527 edges) from BioGRID and 4,364 PSNs constructed from 3D structures from Protein Data Bank (PDB). We use biological process GO terms as protein functional labels for classification, but we only keep all 107 GO terms that annotate at least 20 proteins in our NoN data. For each of the 107 considered GO terms, we employ cross-validation to predict whether each protein is annotated with the given GO term, and subsequently evaluate these predictions across all proteins via the following measures: area under the precision recall curve (AUPR), precision, recall, and F-score.

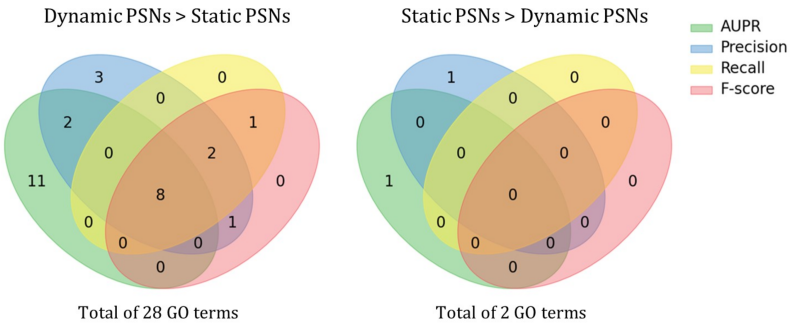

Figure 2: The number of GO terms where a given approach (left: dynamic PSNs; right: static PSNs) performs at least 10% better (absolute improvement) than the other approach, and the given approach's accuracy has a p-value ¡ 0.05, with respect to AUPR, precision, recall, and F-score (and their overlaps).

We find that using dynamic PSNs in the NoN framework exhibits better performance compared to using static PSNs for 91 out of the 107 GO terms for at least one of the four evaluation measures. Of these, the performance of dynamic PSNs exceeds that of static PSNs by at least 10% (absolute improvement) for 62 GO terms. Among these, for 28 GO terms, the performance of dynamic PSNs yields p-values ¡ 0.05. In other words, for the majority of GO terms, the results for using dynamic vs. static PSNs within the NoN framework are closely comparable (within 10%) or random. Yet, when we specifically focus on GO terms where a given approach is superior to the other approach by at least 10% and the former yields p-values ¡ 0.05, we observe the following (Fig. 2). Dynamic PSNs outperform static PSNs in the NoN framework for 28 GO terms with respect to at least one measure. In contrast, static PSNs outperform dynamic PSNs for only 2 GO terms with respect to at least one measure. When considering GO terms at the intersection of all four measures, dynamic PSNs outperform static PSNs for 8 GO terms. In contrast, static PSNs outperform dynamic PSNs for 0 GO terms. So, for GO terms where dynamic or static PSNs yield p-values ¡ 0.05 and one of the approaches is at least 10% better than the other, dynamic PSNs are superior to static PSNs. Of the 8 GO terms for which dynamic PSNs are superior to static PSNs for all four measures, half are biological processes related to (de/auto)phosphorylation, indicating that dynamic PSNs might favor certain types of GO terms over others. We are currently analyzing potential biological implications of these results.

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